

Scalability & Future Plan

Date	1 October 2026
Team ID	6ab501466f290be8a6b4f5d3
Project Name	ComicCraft: AI Comic Story Creator
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Depends on Gemini API quota and rate limits, and older models are retired by Google from time to time.	Generation can fail or slow down when many comics are requested, and model names must be updated.	High
2	Character appearance can vary slightly between panels.	The hero may look different from panel to panel, reducing story consistency.	Medium
3	Runs only locally (localhost) with no saved story history; speech bubbles are not drawn on images.	Users cannot access earlier comics or share the app, and dialogue appears as text outside the image.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single Streamlit app run locally for one user with one API key.	Deploy to the cloud (Docker / Streamlit Cloud), add rate limiting per user and a request queue, and use a higher API quota tier.
2	Data Storage	Generated PDFs are saved in the exports folder and results are kept only in session state.	Store story history, images and PDFs in a database and cloud storage so users can reopen earlier comics.
3	Performance	Panel images are generated one after another, so a long comic takes more time.	Generate panels in parallel (async calls), cache results, and regenerate only the panels that change.
4	Security	API key kept in the .env file and never committed to GitHub.	Move keys to a secrets manager, add user login, validate user input and set usage limits per user.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Draw speech bubbles on the images and use reference images for a consistent hero across panels.	1 - 2 months	More professional-looking comics with consistent characters.
Phase 3	Multi-language story support and saved story history with user accounts.	3 - 4 months	Reaches more users and lets them revisit and share their comics.
Phase 4	Cloud deployment with parallel image generation, caching and usage limits.	5 - 6 months	Supports more users at the same time with faster generation.